LESSON 8

ANOVA

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Index

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Decomposition of variability

Variability definition

The total spread of a variable, decomposed into explained (**systematic**) and unexplained (**random**) components.

Total variability in the data	=	Variation due to underlying factors	+	Residual variation within groups
SCF Total sum of squares	=	SCF Factor sum of squares	+	SCR Residual sum pof squares

Total sum of squares (SCT)

Measures the **total variability** in the data:

$$SCT = \sum_{ij} (x_{ij} - \bar{\bar{x}})^2$$

Where $\bar{\bar{x}}$ is the total mean, $i \in [1 \dots I]$ and represents the number of factors and $j \in [1 \dots J]$ are the number of repetitions. $\mathbf{N} = \boldsymbol{i \cdot j}$ and $\boldsymbol{gl_{total} = N - 1}$**.**

Factor sum of squares (SCT)

Measures the variability in the data associated with **the effect of the factor on the mean**:

$$SCF = \sum_i N_i(\bar{x_i} - \bar{\bar{x}})^2$$

Where $\bar{x_i}$ and N_i are the mean and number of repetition for factor i, respectively. And $\boldsymbol{gl_{factor} = I - 1.}$

The **variance** can be estimated as:

$$s^2 = \frac{SCF}{gl_{factor}}$$

Residual sum of squares (SCT)

Measures **internal variability within each factor**, measurement errors, etc:

$$SCR = \sum_{ij} (x_{ij} - \overline{x_i})^2$$

And $\boldsymbol{gl_{residual} = gl_{total} - gl_{factor} = N - I}$

The **variance** can be estimated as:

$$s^2 = \frac{SCR}{gl_{residual}}$$

Fisher's F-test - Definition

Hypothesis testing:

$$\begin{cases} H_0: m_1 = m_2 = \cdots = m_I, \forall i \ \in I \\ H_1: \exists i / m_i \neq m_{i+1} \end{cases}$$

If the null hypothesis is **accepted**:

- The sample means will also be similar to the population mean:

$$\boldsymbol{m_1 = m_2 = \cdots = m_I \ \approx \bar{\bar{x}}}$$

- The total sum of squares will be small

Fisher's F-test - Definition

Mean of squares calculation

$$CM_{factor} = \frac{SCF}{gl_{factor}}$$

$$CM_{residual} = \frac{SCR}{gl_{residual}}$$

F-ratio calculation

$$F_{ratio} = \frac{CM_{factor}}{CM_{residual}}$$

$$F_{ratio} \sim F_{gl_{factor}, gl_{residual}}$$

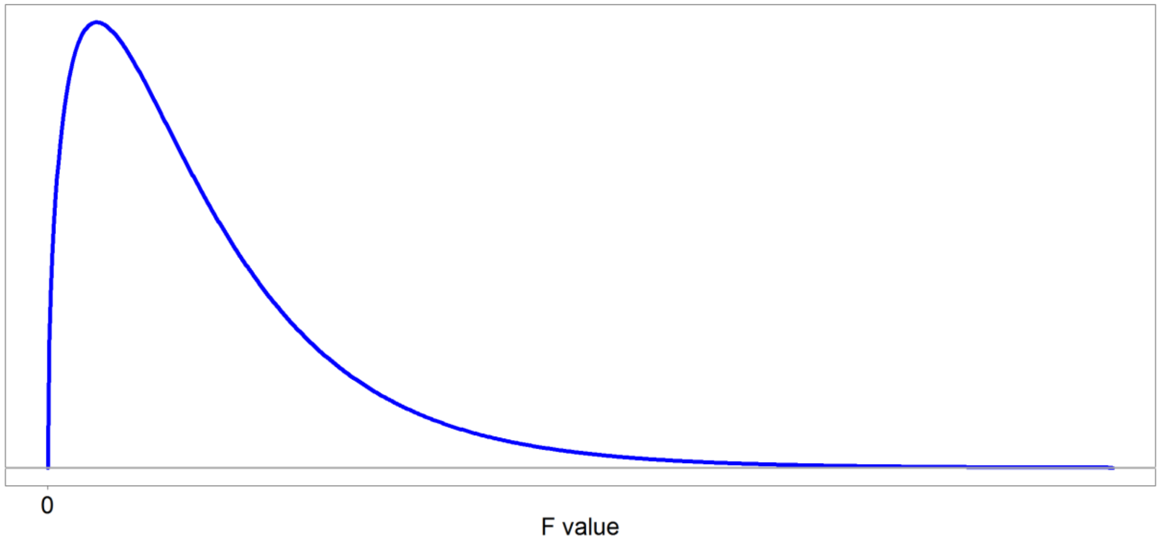
F-Snedecor distribution

$$\boldsymbol{F_{n_1,n_2} = \frac{\frac{\chi^2_{n_1}}{n_1}}{\frac{\chi^2_{n_2}}{n_2}}}$$

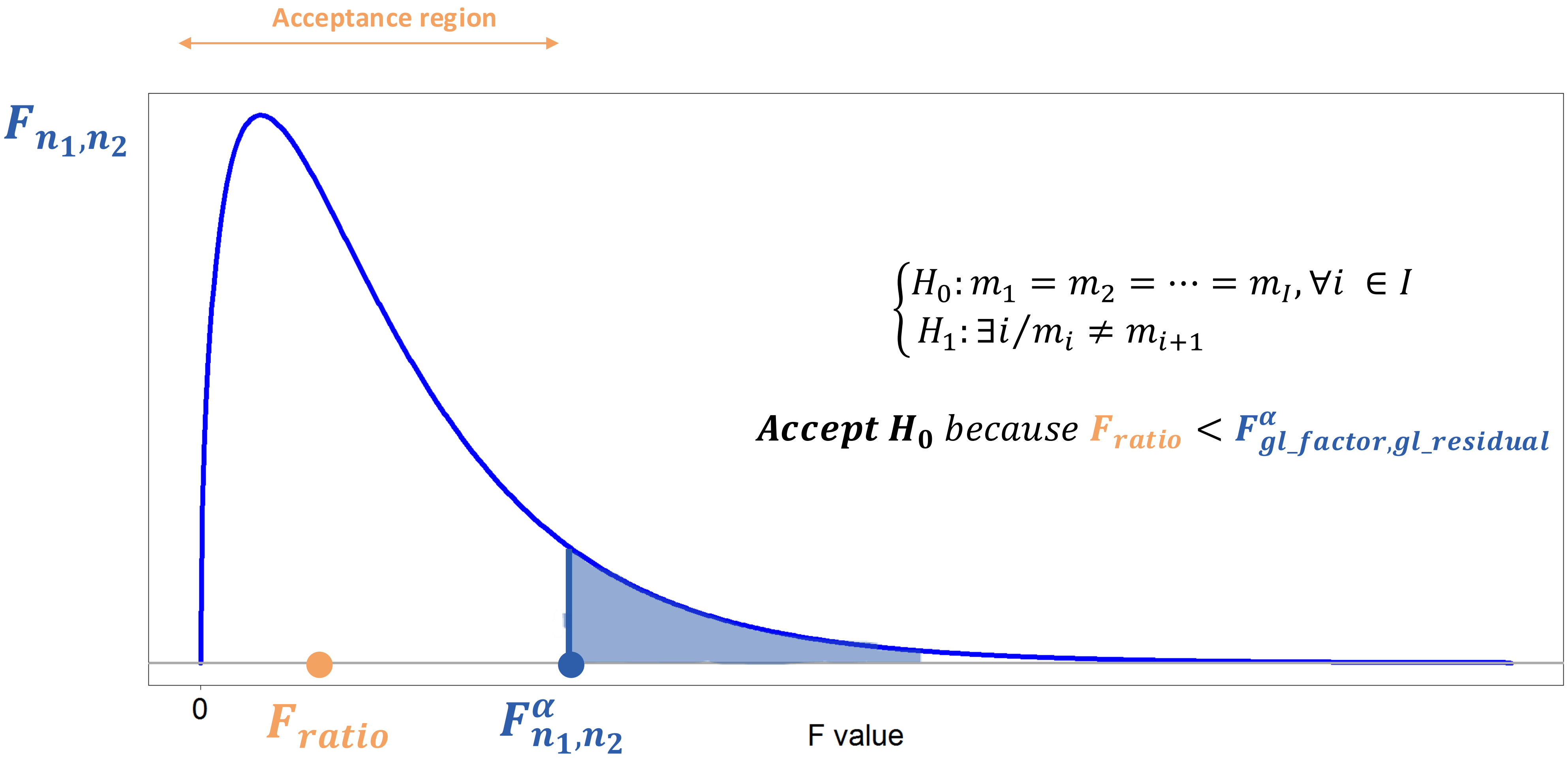
where $\chi^2_{n_1}$ and $\chi^2_{n_2}$ are **independent**

If the two population variances are equal, the ratio is distributed as an F-distribution with $(gl_{factor}, gl_{residual})$ degrees of freedom

An F distribution



F-Snedecor distribution



$$\begin{cases} H_0: m_1 = m_2 = \cdots = m_I, \forall i \in I \\ H_1: \exists i / m_i \neq m_{i+1} \end{cases}$$

$$\boldsymbol{Accept\ H_0}\ because\ F_{ratio} < F^{\alpha}_{gl\_factor,gl\_residual}$$

ANOVA Table

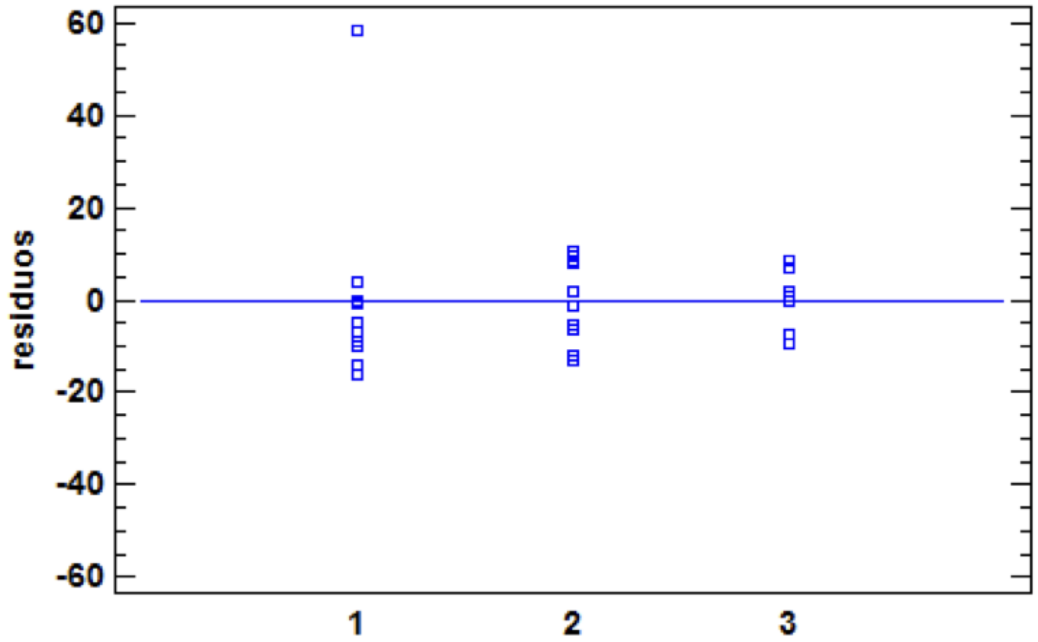
Variation	Sum of squares	Degrees of freedom (gl)	Mean of squares	F-ratio
Total	SCT	gl_{total}	-	
Factor	SCF	gl_{factor}	$CM_{factor} = \frac{SCF}{gl_{factor}}$	$F_{ratio} = \frac{CM_{factor}}{CM_{residual}}$
Residual	SCR	$gl_{residual}$	$CM_{residual} = \frac{SCR}{gl_{residual}}$	

Analysis of residuals

Reflects the impact of all **uncontrolled factors** that may have influenced the result

$$e_{ij} = x_{ij} - \overline{x_i}$$

Enables the detection of anomalous data that may have had a decisive influence on the (erroneous) conclusions of a study.



Analysis of residuals

Basic concepts

- **Homocedasticity:** the variability of residuals is constant across all levels of an independent variable.

- **Heterocedasticity:** the variability of errors or residuals changes across levels of an independent variable.

LSD intervals

By accepting H_1 (there is at least one mi that differs from the others), the aim is to determine which factor i differs from the rest.

$$LSD = \bar{x}_i \pm \frac{\sqrt{2}}{2} \cdot t_{gl_{residual}}^{\alpha/2} \cdot s_{\bar{x}_i}$$

Where $s_{\bar{x}_i} = \sqrt{\frac{CMR}{N_i}}$ and $t_{gl_{residual}}^{\alpha/2}$ is calculated using the t-Student table.

The difference between two means **will be significant** if their LSD intervals **do not overlap.**

Factorial plan

Consider a study designed to investigate the effect of K factors:

- Factor 1 with n_1 variants or levels
- Factor 2 with n_2 variants or levels
- …
- Factor K with n_k variants or levels

This is a special design for the simultaneous study of two or more factors, in which all levels of all factors are combined with one another. In this case:

$$\boldsymbol{n_1 \cdot n_2 \cdot \cdots \cdot n_k} \textbf{ populations would be studied}$$

Factorial plan

The ANOVA for the results of a factorial design is carried out using the same procedures and formulas as those discussed for a single-factor design. **Adding the um of Squares of the Interaction.**

Balanced factorial plan

All possible combinations/populations are studied the same number of times ($\boldsymbol{n_r}$)

- If $\boldsymbol{n_r = 1}$ → It would be a non-replicated balanced factorial design

Multifactorial ANOVA

$$SCF = n_i \sum_i (\bar{x}_i - \bar{\bar{x}})^2 = \frac{\sum_i {T_i}^2}{I \cdot n_r} - SG$$

$$gl_{factor} = I - 1$$

$$SC_{IJ} = n_r \sum_i (\overline{x_{ij}} - \bar{x}_i - \bar{x}_j + \bar{\bar{x}})^2 = \frac{\sum_i {T_{ij}}^2}{n_r} - SG - SC_I - SC_J$$

$$gl_{interaction} = (I-1)(J-1)$$

Multifactorial ANOVA

$$T_{ij} = \sum_{n} x_{ijn_r}$$

$$T_i = \sum_{i} x_{ijn_r}$$

$$T_j = \sum_{j} x_{ijn_r}$$

$$TG = \sum_{i} T_i = \sum_{j} T_j$$

$$SG = \frac{TG^2}{I \cdot J \cdot n_r}$$

Where the Ts represent the sums of the data by row, by column and the totals

Multifactorial ANOVA

$$SCT = \sum_{f}^{F} SC_f + SC_{interaction} + SC_{residual}$$

Where **F** is the number of factors analysed in the Multifactorial ANOVA

Optimal Operating Conditions (OOC)

When you have several factors (I, J, K, …), each with different levels, the goals of multifactor ANOVA are:

- Identifying which factors significantly affect the response.
- Whether there are interactions between factors.

The **OCC** refer to the specific levels of all factors that optimize the response, based on the statistical analysis.

Case study 1

An engine manufacturer has two suppliers for the crankshafts it machines. A third supplier offers slightly more expensive crankshafts, citing their superior dynamic properties, specifically, that their dynamic balance (the number of grams of material that must be removed to ensure the part's centre of gravity aligns with the axis of rotation) is lower.

The factory decides to conduct a test comparing 10 crankshafts from the new supplier (S=1) with 10 from each of its two traditional suppliers (S=2 and S=3).

Based on the results obtained, the aim is to investigate whether there is sufficient evidence of the superiority of the new supplier's crankshafts to justify switching suppliers despite the slightly higher price.

Case study 1

Supplier 1	Supplier 2	Supplier 3
23	35	50
28	36	43
21	29	36
27	40	34
95	43	45
41	49	52
37	51	52
30	28	43
32	50	44
36	52	34

I = 3, $i \in [1, 2, 3]$

$$\overline{x_1} = 37.0$$

$$\overline{x_2} = 41.3$$

$$\overline{x_3} = 43.3$$

$$\overline{\overline{x}} = 40.53$$

J = 10, $j \in [1, 2, \ldots, 10]$

$$N = I \cdot J = 10 \cdot 3 = 30$$

Case study 1

$$SCT = \sum_{ij} (x_{ij} - \bar{\bar{x}})^2$$

$$SCT = (23 - 40.53)^2 + (35 - 40.53)^2 + \cdots + (34 - 40.53)^2 = \mathbf{5465}$$

$$gl_{total} = N - 1 = 30 - 1 = \mathbf{29}$$

Case study 1

$$SCF = \sum_i N_i(\overline{x_i} - \overline{\overline{x}})^2$$

$$SCR = 10 \cdot (37 - 40.53)^2 + 10 \cdot (41.3 - 40.53)^2 + (43.3 - 40.53)^2 = \mathbf{207}$$

$$gl_{factor} = 3 - 1 = \mathbf{2}$$

Case study 1

$$SCR = \sum_{ij} (x_{ij} - \overline{x_i})^2$$

$$SCR = (23-37)^2 + (35-41.3)^2 + (50-43.3)^2 + \cdots + (36-37)^2 + (52-41.3)^2 + (34-43.3)^2 = \mathbf{5258}$$

$$gl_{residual} = 29 - 2 = 30 \ - 3 = \mathbf{27}$$

Case study 1

Mean of squares calculation

$$CM_{factor} = \frac{207}{2} = 103.5$$

$$CM_{residual} = \frac{5258}{27} = 194.7$$

F-ratio calculation

$$F_{ratio} = \frac{103.5}{194.7} = 0.532$$

Case study 1

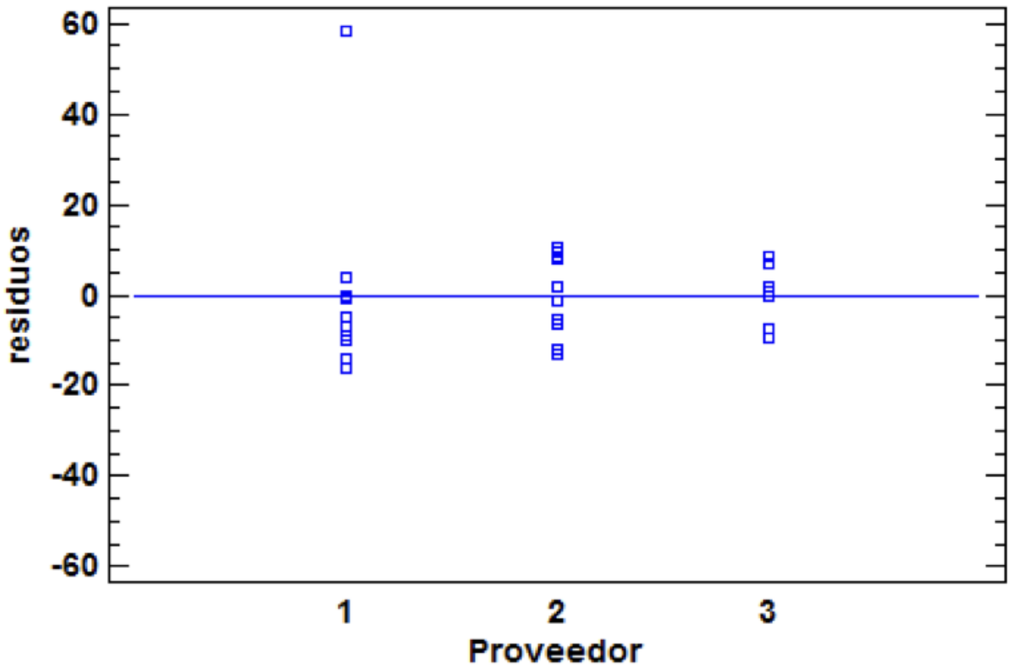
Variation	Sum of squares	Degrees of freedom (gl)	Mean of squares	F-ratio
Total	$SCT = 5465$	$gl_{total} = 29$	-	
Factor	$SCF = 207$	$gl_{factor} = 2$	$CM_{factor} = 103.5$	$F_{ratio} = 0.532$
Residual	$SCR = 5258$	$gl_{residual} = 27$	$CM_{residual} = 194.7$	

$$\boldsymbol{F_{2,29} = 3.32}$$

$F_{2,29} > 0.532 \rightarrow$ $\boldsymbol{H_0}$ **is accepted**, there are not statistical differences between suppliers

Case study 1

The graph clearly shows the presence of an outlier for supplier 1, as indicated by a residual value of almost 60, which is much higher than any of the other values obtained.



Case study 1

New ANOVA after correcting the outlier:

Variation	Sum of squares	Degrees of freedom (gl)	Mean of squares	F-ratio
Total	$SCT = 2409,47$	$gl_{total} = 29$	-	
Factor	$SCF = 871,26$	$gl_{factor} = 2$	$CM_{factor} = 435.63$	$F_{ratio} = 7,65$
Residual	$SCR = 1538,20$	$gl_{residual} = 27$	$CM_{residual} = 56.09$	

$$\boldsymbol{F_{2,27} = 3.32}$$

$F_{2,27} < 7.65 \rightarrow$ $\boldsymbol{H_0}$ **is rejected**

Case study 1

$$\boldsymbol{LSD_1} = 37 \pm \frac{\sqrt{2}}{2} \cdot t_{gl_{27}}^{0.05/2} \cdot \sqrt{\frac{56.09}{10}} = 37 \pm 1.086 = [33.564\,,40.436]$$

$$\boldsymbol{LSD_2} = 41.3 \pm \frac{\sqrt{2}}{2} \cdot t_{gl_{27}}^{0.05/2} \cdot \sqrt{\frac{56.09}{10}} = 41.3 \pm 1.086 = [39.317\,,43.283]$$

$$\boldsymbol{LSD_3} = 43.3 \ \pm \frac{\sqrt{2}}{2} \cdot t_{gl_{27}}^{0.05/2} \cdot \sqrt{\frac{56.09}{10}} = 43.3 \pm 1.086 = [41.317\,,45.283]$$

There is a **significant difference between the mean for Supplier 1 and those for the other two suppliers**; however, the difference between the latter two is not significant.

Case study 2

We wish to determine the effects of the type of catalyst used in the synthesis of a chemical product and the supplier of the raw material on the quantity of product obtained per hour.

To this end, **two types of catalyst (A and B)** are compared and **three different suppliers (1, 2 and 3)** are tested. Twelve tests are carried out, two for each of the possible combinations of factors (6 combinations).

- Factor 1: Type of catalyst with 2 variants (A and B)
- Factor 2: Type of supplier with 3 variants (1, 2 and 3)

In this case, $n_1 \cdot n_2 = 2 \cdot 3 = 6$ different populations are studied, and 2 tests are carried out for each population ($n_r = 2$). In total, there are **6 · 2 = 12 tests**

Case study 2: Case A

Let's imagine we want to investigate whether the type of supplier and the type of catalyst influence the production of the chemical; the results are shown in the table below

Supplier (S)

Catalyst (C)

	1	2	3	
A	20 20	20 20	20 20	$\overline{x_A} = 20$
B	20 20	20 20	20 20	$\overline{x_B} = 20$
	$\overline{x_1} = 20$	$\overline{x_2} = 20$	$\overline{x_3} = 20$	$\bar{\bar{x}} = 20$

$$SCT = \sum (x_{ijn_r} - \bar{\bar{x}})^2 = \sum (20 - 20)^2 = 0$$

Neither the type of supplier nor the type of catalyst affects the quantity of the product**. If there is no variability, nothing affects it.**

Case study 2: Case B

The results have now changed slightly:

Supplier (S)

Catalyst (C)	1	2	3	
A	20 20	20 20	20 20	$\overline{x_A} = 20$
B	30 30	30 30	30 30	$\overline{x_B} = 30$
	$\overline{x_1} = 25$	$\overline{x_2} = 25$	$\overline{x_3} = 25$	$\bar{\bar{x}} = 25$

$$SCT = \sum (x_{ijn_r} - \bar{\bar{x}})^2 = \sum (20 - 25)^2 + \sum (30 - 25)^2 = 12 \cdot 25 = 300 = SC_C$$

The variability in the mean is due solely to the effect of the catalyst type. **There is no effect from the supplier.**

Case study 2: Case C

The results are now changing slightly once again:

		Supplier (S) 1	2	3	
Catalyst (C)	A	20 20	25 25	30 30	$\overline{x_A} = 25$
	B	30 30	35 35	40 40	$\overline{x_B} = 35$
		$\overline{x_1} = 25$	$\overline{x_2} = 30$	$\overline{x_3} = 35$	$\bar{\bar{x}} = 30$

$$SCT = \sum (x_{ijn_r} - \bar{\bar{x}})^2 = 2 \cdot (20 - 30)^2 + 2 \cdot (25 - 30)^2 + \cdots + 2 \cdot (40 - 30)^2 = \mathbf{500}$$

$$SCT = 500 = SC_C + SC_S$$

The variability is due both to the effect of the catalyst type and to the effect of the supplier type. There is no interaction between the factors (the difference between A and B is the same across all three levels of Supplier)

Case study 2: Case D

Supplier (S)

Catalyst (C)	1	2	3	
A	20 20	25 25	30 30	$\overline{x_A} = 25$
B	30 30	40 40	50 50	$\overline{x_B} = 40$
	$\overline{x_1} = 25$	$\overline{x_2} = 32{,}5$	$\overline{x_3} = 40$	$\bar{\bar{x}} = 32{,}5$

$$SCT = \sum (x_{ijn_r} - \bar{\bar{x}})^2 = 2 \cdot (20 - 32.5)^2 + 2 \cdot (25 - 32.5)^2 + \cdots + 2 \cdot (50 - 32.5)^2 = \mathbf{1175}$$

$$SCT = 1175 = SC_C + SC_S + SC_{interaction}$$

There is variability due both to the individual effects of the factors and to their interaction. This interaction is evident in the fact that the difference between catalysts A and B varies for each supplier variant, with this difference increasing as the supplier code increases (1,2,3)

Case study 2: Case E (real case)

Supplier (S)

Catalyst (C)	1	2	3	
A	19 21	27 24	28 32	$\overline{x_A} = 25{,}167$
B	30 31	42 39	47 51	$\overline{x_B} = 40$
	$\overline{x_1} = 25{,}25$	$\overline{x_2} = 33$	$\overline{x_3} = 39{,}5$	$\bar{\bar{x}} = 32{,}583$

$$SCT = \sum (x_{ijn_r} - \bar{\bar{x}})^2 = (19 - 32.583)^2 + (27 - 32.583)^2 + \cdots + (51 - 32.583)^2 = \mathbf{1130{,}92}$$

$$SCT = 1130{,}92 = SC_C + SC_S + SC_{interaction} + SC_{residuals}$$

The effects of the factors are partially masked by the residual variability caused by uncontrolled factors. **Data pairs with the same Catalyst type and the same Supplier do not yield exactly the same Product results.**

Exercise 1

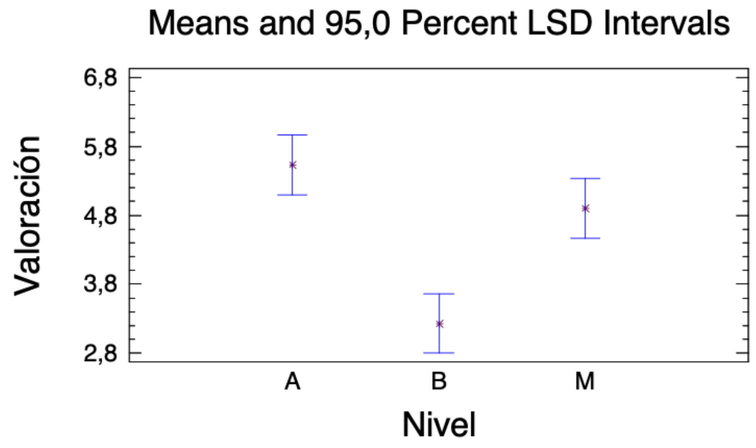
Data has been collected on the ratings given to a political leader in two different cities, 1 and 2 (Factor A), each of which is divided into three neighbourhoods according to income level (High, Medium and Low) (Factor B). It appears that the most appropriate method for determining whether these two factors influence these ratings is an analysis of variance:

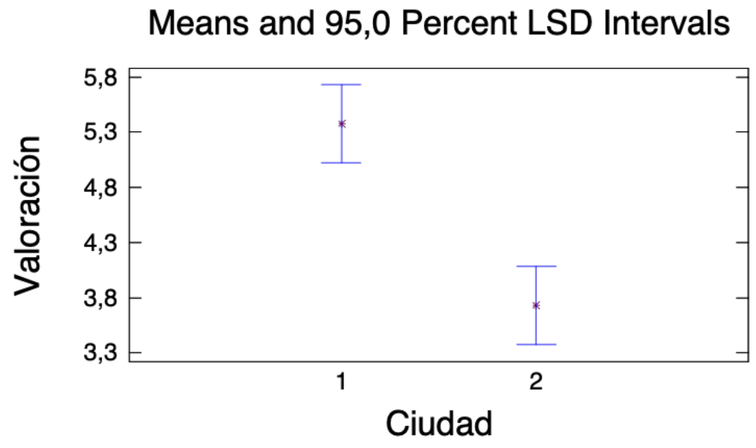
Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
MAIN EFFECTS					
A:Ciudad	60,8444	1	60,8444	21,37	0,0000
B:Nivel	84,6889	2	42,3444	14,87	0,0000
INTERACTIONS					
AB	29,4889	2	14,7444	5,18	0,0076
RESIDUAL	239,2	84	2,84762		
TOTAL (CORRECTED)	414,222	89			

Exercise 1

a) What conclusions can be drawn

b) In light of the following graphs, and taking into account the conclusions reached in the previous section, identify which city has the highest rating for the political leader and what their purchasing power is, explaining which graphs you have used to reach these conclusions and why.





Exercise 2

In a study aimed at improving the top speed achieved by a new racing car prototype, two aerodynamic configurations (A1 and A2) and two fuel variants (CB1 and CB2) will be tested.

The response variable to be maximised is the speed reached at the end of the main straight at the Montmeló circuit. To carry out the experiment, 64 speed measurements are taken, with 16 tests for each of the possible treatments.

The results of the study have been analysed using an analysis of variance, the results of which are shown in the table.

Exercise 2

NOTA: $F_{1,63}$**(5%)**=3.99 ; $F_{1,63}$**(2.5%)**= 5.272 ; $F_{2,63}$**(5%)**=3.142 ; $F_{2,63}$**(2.5%)**=3.914 ; $F_{1,60}$**(5%)**=4 ; $F_{1,60}$**(2.5%)**= 5.285 ; $F_{2,60}$**(5%)**=3.15 ; $F_{2,60}$**(2.5%)**= 3.925

Análisis de Varianza para Velocidad - Suma de Cuadrados Tipo III

Fuente	*Suma de Cuadrados*	*Gl*	*Cuadrado Medio*	*Razón-F*	*Tabla F*
EFECTOS PRINCIPALES					
A:Combustible				167,40	
B:Config Aerodinámica	484,0				
INTERACCIONES					
AB	0,5625				
RESIDUOS			2,51042		
TOTAL (CORREGIDO)	1055,44				

Tabla de Medias por Mínimos Cuadrados

Nivel	*Casos*	*Media*
MEDIA GLOBAL	64	302,406
Combustible		
CB1	32	304,969
CB2	32	299,844
Config Aerodinámica		
A1	32	299,656
A2	32	305,156
Combustible por Config Aerodinámica		
CB1;A1	16	302,125
CB1;A2	16	307,813
CB2;A1	16	297,188
CB2;A2	16	302,5

Análisis de Varianza para RESIDUOS^2 - Suma de Cuadrados Tipo III

Fuente	*Suma de Cuadrados*	*Gl*	*Cuadrado Medio*	*Razón-F*	*Tabla F*
EFECTOS PRINCIPALES					
A:Combustible	2,9541			0,45	
B:Config Aerodinámica	0,610352			0,09	
INTERACCIONES					
AB	0,0119629			0,00	
RESIDUOS	397,576				
TOTAL (CORREGIDO)	401,153				

Exercise 2

a) Based on the results obtained, can the assumption of homoscedasticity be considered to hold? Explain what this assumption entails.

b) Identify the treatment that leads to the maximisation of the speed reached at the end of the straight.

c) Calculate the probability that a vehicle under these conditions will reach a speed between 307 and 308 km/h at the end of the main straight at Montmeló.

Exercise 3

An electrical engineer is investigating a plasma etching process used in the manufacture of semiconductors. The aim is to study the effects of two factors: the flow rate of C2F6 gas, with two levels, and the power applied to the cathode, with three levels. The response is the etching rate. Two experimental replicates are carried out for the etching rate for each of the treatments. The design is a completely randomised design, with the factor levels in ascending order, where level 1 is the lowest and level 3 the highest.

The ANOVA tables resulting from the study are presented (INCOMPLETE) below, along with the corresponding tables of means.

Exercise 3

Análisis de Varianza para Velocidad de grabado - Suma de Cuadrados Tipo III

Fuente	*Suma de Cuadrados*	*Gl*	*Cuadrado Medio*	*Razón-F*	*Valor-P*
EFECTOS PRINCIPALES					
A:Cantidad de flujo C2F6	4181,33				
B:Potencia Suministrada	236641,				
INTERACCIONES					
AB					
RESIDUOS	5717,0				
TOTAL (CORREGIDO)	248628,				

Análisis de Varianza para RESIDUOS^2 - Suma de Cuadrados Tipo III

Fuente	*Suma de Cuadrados*	*Gl*	*Cuadrado Medio*	*Razón-F*	*Valor-P*
EFECTOS PRINCIPALES					
A:Cantidad de flujo C2F6					0,0000
B:Potencia Suministrada	461368,				0,0000
INTERACCIONES					
AB	948626,				0,0000
RESIDUOS	3,49246E-10				
TOTAL (CORREGIDO)	1,92916E6				

Exercise 3

Tabla de Medias por Mínimos Cuadrados para Velocidad de grabado con intervalos de confianza del 95,0%

Nivel	*Casos*	*Media*	*Error Est.*	*Límite Inferior*	*Límite Superior*
MEDIA GLOBAL		517,167			
Cantidad de flujo C2F6					
1		498,5	12,6018	467,664	529,336
2		535,833	12,6018	504,998	566,669
Potencia Suministrada					
1		361,0	15,434	323,234	398,766
2		489,0	15,434	451,234	526,766
3		701,5	15,434	663,734	739,266
Cantidad de flujo C2F6 por Potencia Suministrada					
1,1		324,0	21,827	270,591	377,409
1,2		476,5	21,827	423,091	529,909
1,3		695,0	21,827	641,591	748,409
2,1		398,0	21,827	344,591	451,409
2,2		501,5	21,827	448,091	554,909
2,3		708,0	21,827	654,591	761,409

Exercise 3

Tabla de Medias por Mínimos Cuadrados para RESIDUOS^2 con intervalos de confianza del 95,0%

			Error	*Límite*	*Límite*
Nivel	*Casos*	*Media*	*Est.*	*Inferior*	*Superior*
MEDIA GLOBAL		476,417			
Cantidad de flujo C2F6					
1		684,417		684,417	684,417
2		268,417		268,417	268,417
Potencia Suministrada					
1		732,5	0,0000038147	732,5	732,5
2		256,25	0,0000038147	256,25	256,25
3		440,5	0,0000038147	440,5	440,5
Cantidad de flujo C2F6 por Potencia Suministrada					
1,1		1296,0	0,0000053948	1296,0	1296,0
1,2		132,25	0,0000053948	132,25	132,25
1,3		625,0	0,0000053948	625,0	625,0
2,1		169,0	0,0000053948	169,0	169,0
2,2		380,25	0,0000053948	380,25	380,25
2,3		256,0	0,0000053948	256,0	256,0

Exercise 3

For a Type I risk of 5%:

Complete the necessary ANOVA table to indicate the statistical significance of the effect of each of the factors involved, including the interaction, on the mean value of the engraving speed.

What does it mean for the interaction to be (or not to be) statistically significant? Determine the optimal operating conditions (OOC) that maximise the mean value of the recording speed.

In view of the tables above, what can be deduced regarding the variability of the recording speed under the OOC?